

1. Introduction

For many years, the University of Eswatini, specifically the Student Finance Office, has been using face to face methods for student financial clearance process. We, as the initiators of this project have had a hand-first experience of this process and have suffered the high monetary costs and time consumption that are the results of this issue. This can be experienced particularly during semester holidays when students are away from the campus. Even in the case that the institution is open, there are still long queues and delays which lead to an overall inefficient process. This drew our attention and prompted us develop a system that will mitigate this issue by digitizing and automating the whole clearing process, hence saving time of students and providing a seamless user experience. This will also benefit the institution in terms of aligning this procedure with the digital nature of its other processes like applications and registration.

2. User / Client Involvement

Although most of the project and end-product will be attended to by the development group, the users also play an essential role in the functionality of the system but providing the information that will trigger the process. The information that the students must provide is:

- Student Identification Number
- Student passwords
- Proof of payment
- Financial Statement

3. Risks

Like any project, there are always risks that may emerge. It is important to identify these threats to ensure project success and usability of the system. Here are the possible risks and mitigation strategies for each:

- **Personality theft:** students may gain unauthorized access to other individuals personal information like results and financial records by copying the login details of others.

Mitigation: Allow students to create strong passwords composed of letters, digits, characters and symbols and also hide passwords so that they can't be copied.

- **Document Fraud:** Students may upload fraudulent payment receipts and financial statements.

Mitigation: High resolution verification standards will be used to check the authenticity of documents.

- **System Overload:** The students may overload the system with many documents which may lead to redundancy.

Mitigation: Allow the students to upload a specific number of documents. There must also be the size limit of each document and allow for compressed documents if necessary. Also, create clear labels that inform the users exactly which documents are needed and which are not.

4. Standards, guidelines and procedures

4.1 Standards

The development will follow IEEE Software Engineering Standards. The project must comply with the Eswatini Data protection Act and UNESWA ICT Governance Framework. All code will be stored in a private repository following a strict Git-flow branching model.

4.2 Procedures and Guidelines

Planning

- Engage with the institution to find out if they need the service. If they do, then an agreement will be reached.
- Conduct user analysis using questionnaires to get the views of students on the project so that we deliver what the users need.
- Decide what platforms to use and programming languages to write the code.

- Decide on the duration of the project and calculate the expected budget.

Design

- After getting a clear description of the project from the project manager and organizer, which include the assessed user needs, the system will be designed to encompass all of these aspects. The design will include the organization of the content, and vision cues for the user interface.

Implementation

- A programming language and IDE will be chosen such that it aligns with the one that the institution uses and promises high quality of the system.
- The code will be written to materialize the design specified in Design Specification document and as result, we will have a functional system.

Testing

- Find random users (students) to test the system and determine if it performs as intended. The user feedback must be collected and recorded as they navigate the system so as to use it to fix or improve the system.

Evaluation and Deployment

- Submit the finished project to the UNESWA ICT committee so that they can evaluate it, and if they accept it then we deploy the system.

5. Organization of Project

The system will be designed to be synchronized with the e-learning platform, Moodle under the Student Services section. The system will also be linked to the database of the Student Finance Office.

Below are the roles of each key personnel:

- Project Manager: scheduling and resource allocation
- Software Designer: creates the visual representation and blueprint of the system
- Software Architect: labels the components of the system and defines clear navigation paths

- Software Developer: codes the actual system to make it live and functional.

6. Project Phases

The life cycle of the project will be as follows: Planning, Design, Implementation, Testing, Evaluation and Deployment. This life cycle will be implemented using the Incremental Development model because it can handle changes and it allows user feedback.

7. Requirement Analysis and Design

The requirements and design of the system will be decided with the assistance of the following:

- Interviews: a face to face interview with clients and users will be conducted to learn about their preferences. The tools that will be used will include interview forms.
- Design through Persona: A persona (a fake person with the attributes of the user) will be used to ensure that we design to accommodate the needs of different users.

8. Implementation

- Programming Languages such as Javascript, C++ and Python will be used. The most suitable will be chosen because of its alignment with the Moodle platform.
- IDEs such as Google Colab, Visual Studio Code, Git & Github will be utilized.
- Databases such as MySQL, PostgreSQL and MongoDB will also be used to store records

9. Testing

The testing process will be conducted on a simulated database environment mirroring the University's database. The order of the testing process is as follows: Component testing, Integration testing, System Security Audit, Stress testing by simulating up to 5000 concurrent student logins to ensure system stability.

10. Resources

In order for the project to be success, there are some essential resources that must be at our disposal. These are: Computers, Internet connection, Software platforms and Financial support.

11. Quality Assurance

The University IT Department will evaluate the system by allowing some students to try and login to the system and conducting the whole process to see if it performs as expected.

12. Changes

- Every phase of the development process will be tested, evaluated and will be changed and updated according.
- As users interact with the system, they will give feedback which will be used to update the system.

Task	Task Name	Duration (Days)	Start Date	End Date	Cost Per Day (E)	Predecessor
A	Contract signing	3	4 May	6 May	150	-

B	User- analysis	7	7 May	13 May	250	A
C	System Design	10	14 May	23 May	300	A, B
D	Implementati on	15	24 May	7 June	200	B
E	System Testing	7	8 June	14 June	500	D
F	System Evaluation and Deployment	14	15 June	28 June	500	E

Table

Gantt Chart

